



UNIT –II

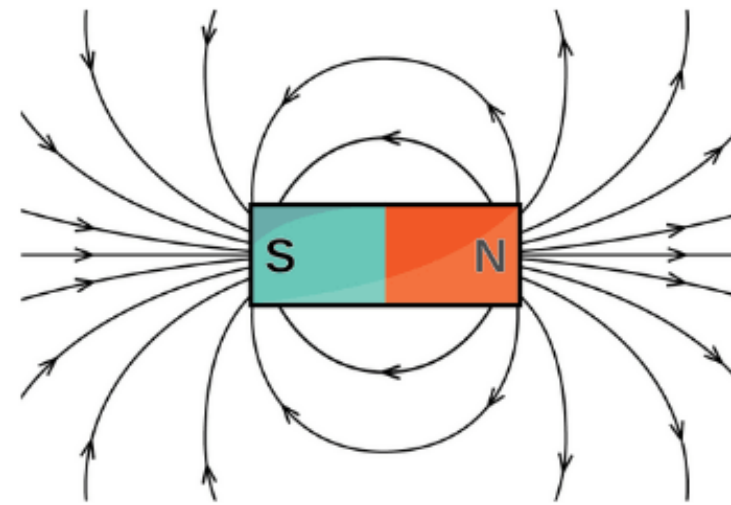
Magnetic Materials

Dr. Ashish Bodhaye
Assistant Professor

**St. Vincent Pallotti College of
Engineering & Technology,
Nagpur**

Magnetic field

- The space around a magnet where its magnetic influence is experienced is known as magnetic field.
- Lode stone or magnetite is a natural magnet.
- In 1819, a Danish Physicist Oersted, discussed that a current carrying conductor produces a magnetic field around it.
- The nature of magnetic field produced is same as that of permanent magnet.



Magnetic field (H):

When the material is kept in magnetic field, the field is called magnetizing field. The strength of magnetic field is called as intensity and it is denoted by H. The unit of H is Ampere per meter (A/m).

Magnetization (M):

The magnetic moment per unit volume developed inside a solid is called magnetization and it is denoted by M. Its unit is (A/m) Ampere/meter.

$$M \propto H \quad \Rightarrow M = \chi H$$

Where χ is magnetic susceptibility which is a constant of proportionality.

Magnetic susceptibility (χ):

Magnetic susceptibility of a material is the magnetization produced in the material per unit applied magnetic field. It is the case with which the material can be magnetized.

$$\chi = \frac{M}{H} \text{-----}(2)$$

Magnetic Induction (B):

Magnetic field is represented by lines of magnetic induction. It can be described by magnetic field strength (H) or magnetic induction B. Magnetic induction is also called as magnetic flux density.

Lines of magnetic induction is nothing but a flux. The number of field lines passing through a unit area of cross-section is called as magnetic flux density.

$$B = \frac{\text{Magnetic flux}}{\text{Area}} = \frac{\phi}{A} \text{-----}(3)$$

Its unit is weber per square meter (W/m²) or Tesla (T). Its CGS unit is Gauss. 1G=10⁻⁴ T.

Relation between B and H:

If a material is kept in magnetic field, magnetizing induction can arise due to magnetic field (H) and magnetization M.

$$B = \mu_0(H + M) \text{-----(4)}$$

Where μ_0 is known as permeability of free space. $\mu_0 = 4\pi \times 10^{-7}$ Henry/meter.

From eqn. (1) and (4) we get

$$B = \mu_0(H + \chi H)$$

$$B = \mu_0(1 + \chi)H$$

$$B = \mu H \text{-----(5)}$$

Where μ is absolute permeability of the medium and $\mu = \mu_0(1 + \chi)$
 χ and μ are in general a tensor. (a tensor quantity is a physical quantity which has no specific direction but different values in different directions). But for isotropic medium they are scalars.

For free space $M = 0$, hence eqn. (4) becomes

$$B = \mu_0.H \text{-----(6)}$$

Absolute permeability (μ):

When a magnetic material is placed in a magnetic field, the field lines are redistributed. These lines tend to pass through the material. Absolute permeability is a measure of the degree of which the field lines penetrate or permeate the material. The ratio of magnetic induction (B) to the magnetizing field (H) is called as absolute permeability.

$$\mu = \frac{B}{H} \text{-----(7)}$$

Unit of μ is Henry per meter (H/m).

Relative permeability (μ_r):

The ratio of absolute permeability of the material to the permeability of free space is called as relative permeability and is expressed as

$$\mu_r = \frac{\mu}{\mu_0} \text{----- (8)}$$

μ_r is a pure number and has no unit. For air or vacuum $\mu_r = 1$.

Relation between μ_r and χ :

We have $\mu = \mu_0(1 + \chi)$ -----(9)

From eqn. (8)

$$\mu_r = \frac{\mu}{\mu_0} \Rightarrow \mu_r \cdot \mu_0 = \mu \text{ -----(10)}$$

Comparing eqn. (9) & (10) we get

$$\mu_r = (1 + \chi)$$

This equation is the relation between susceptibility and permeability.

Magnetic dipole moment (μ_m):

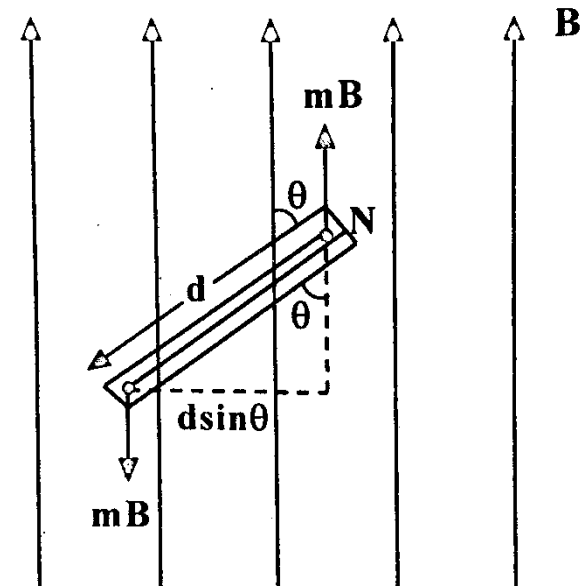
If a small bar magnet is freely suspended in uniform magnetic field of induction B . it experiences a torque which tends to align the magnet in the direction of field. The force acting on each pole is ' mB '. The forces acting on each pole act in opposite directions and constitute a couple as shown in fig.

Magnetic dipole moment is a product of pole strength and perpendicular distance between the poles.

If θ is the angle that a bar magnet makes with B then, the torque τ is given by

$$\tau = \mu_m \cdot B \cdot \sin\theta,$$

Where $\mu_m = m \times d$ and is called as magnetic dipole moment. It is a vector quantity and its direction is from south pole to north pole.



Origin of magnetization:

Magnetic properties arise due to electrons undergoing different motions in the atoms. This motion gives rise to magnetic dipole moment which is responsible for the magnetic properties of materials. Magnetic dipole moment arises due to three reasons.

I. Orbital motion of electron:

Atom consists of central nucleus and electrons revolve around it in specific orbit. Each electron orbit is equivalent to a tiny current loop and behaves as a tiny magnet having magnetic dipole moment. Total orbital magnetic moment is the sum of orbital magnetic moments of individual electrons.

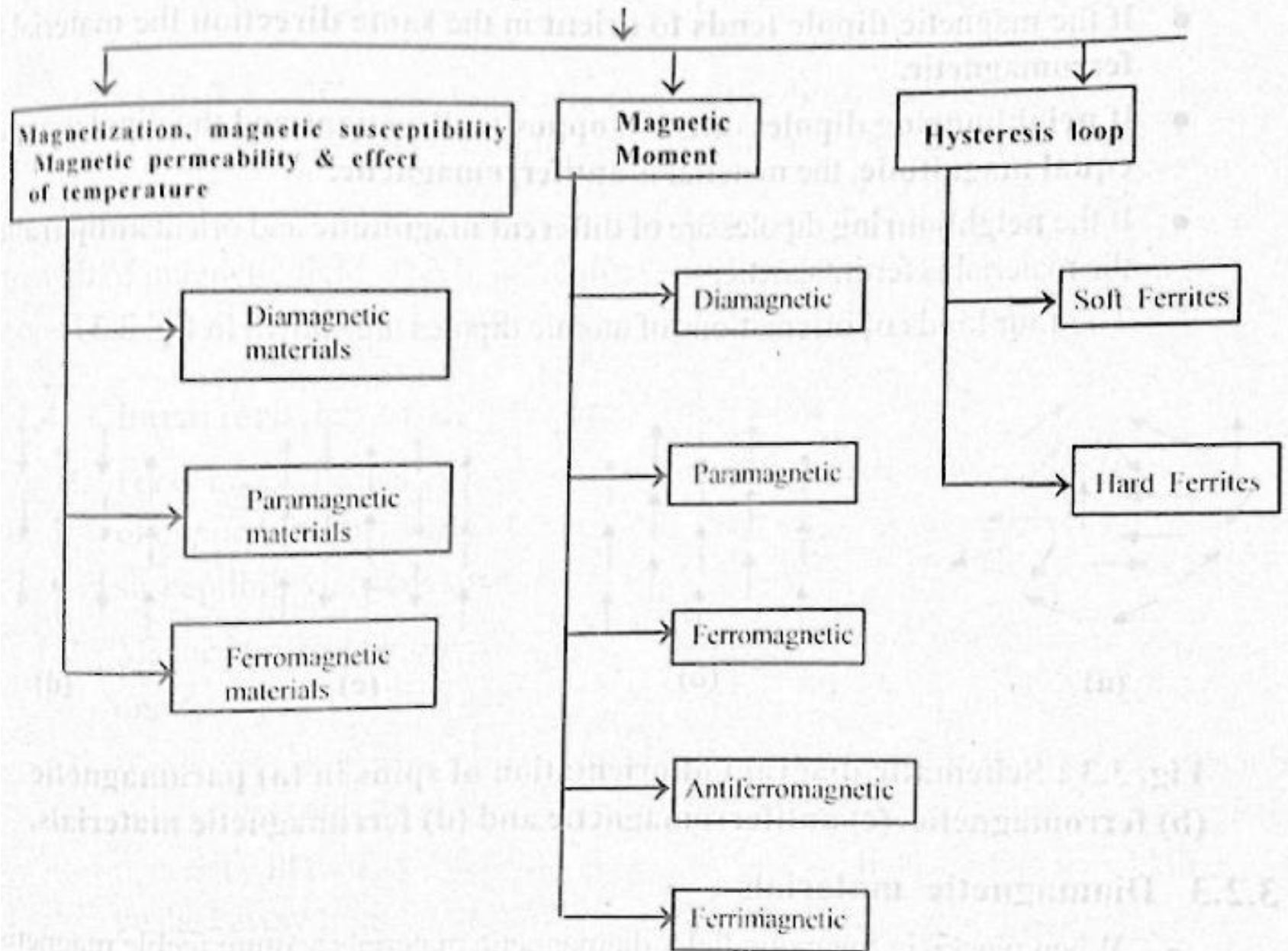
II. Electron spin:

Each electron spins about the axis through itself which gives rise to magnetic dipole moment.

III. Nuclear spin:

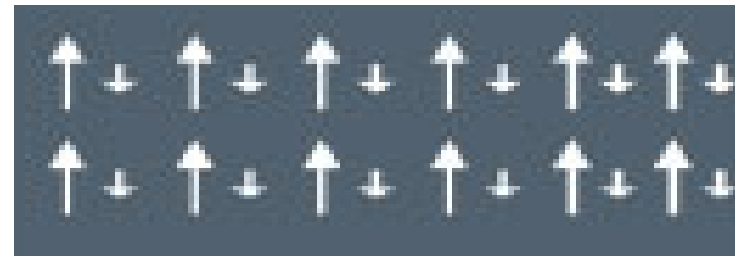
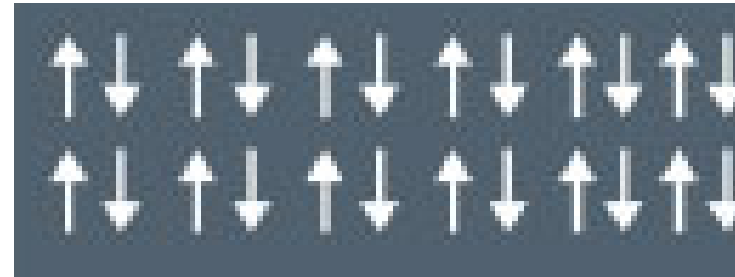
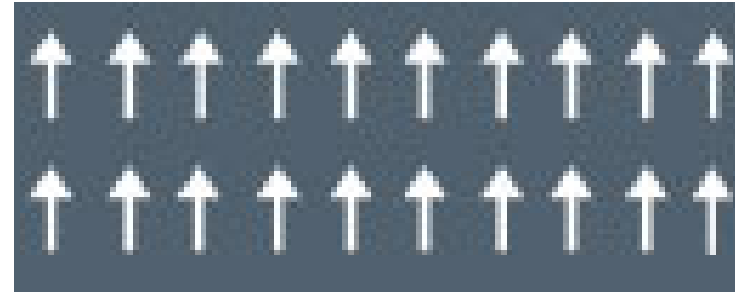
Nuclear spin also gives rise to magnetic moment of atom. Its value is $1/2000$ times as that of magnetic moment of electron. Hence this spin is neglected. Thus a major contribution comes from spin of unpaired valence electrons and orbital motion of an electron.

Magnetic materials



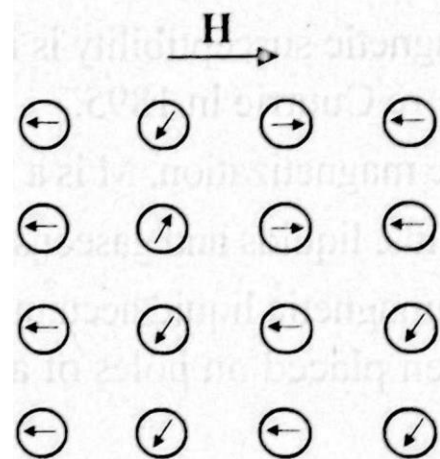
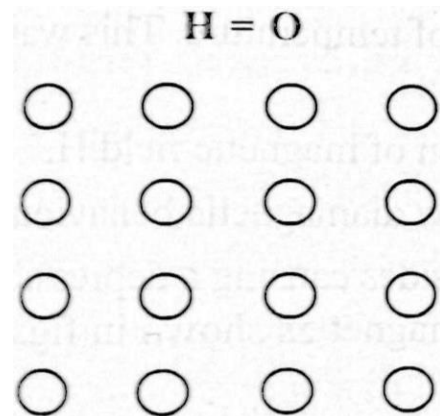
- On the basis of magnitude and sign of relative permeability (μ_r) magnetic materials are divided into three groups:
 1. Diamagnetic materials $\mu_r < 1$
 2. Paramagnetic materials $\mu_r > 1$
 3. Ferromagnetic materials $\mu_r \gg 1$.
- On the basis of magnetic dipole moment the magnetic materials are divided into two groups. One group which do not possess permanent magnetic dipole moment and another group which carry permanent magnetic dipole moment.
- First group is called diamagnetic materials consisting of atoms with zero magnetic dipole moment.
- The second group is divided into four types:
 - 1) Paramagnetic materials
 - 2) Ferromagnetic materials
 - 3) Antiferromagnetic
 - 4) Ferrimagnetic materials.

- If the interaction between the atomic magnetic dipoles is negligible the material is **paramagnetic**.
- If the magnetic dipole tends to orient in the same direction the material is **ferromagnetic**.
- If neighboring dipoles orient in opposite directions and the dipoles are of equal magnitude, the material is **antiferromagnetic**.
- If the neighboring dipoles are of different magnitude and orient antiparallel, the material is **ferrimagnetic**.
- The four kinds of orientations of atomic dipoles are shown in above fig.



Diamagnetic materials:

- When placed in magnetic field, diamagnetic materials acquire feeble magnetism in a direction opposite to that of applied magnetic field. e.g.: Hydrogen. Air, Water, Gold, Silver and Bismuth.
- The materials having atoms consisting of even number of electrons, exhibit diamagnetism. The electron of such atoms are paired. Electrons have orbital as well as spin motions. The magnetic moments cancel each other and hence the resultant magnetic moment of the atom is zero.
- When diamagnetic material is placed in external magnetic field, alignment of magnetic dipoles does not occur. External magnetic field modifies the motion of electrons in the orbits. Each electron pair gives rise to a resultant magnetic moment. However, effective magnetic moment is induced in the atom as a whole in a direction opposite to that of the external magnetic field as shown in fig.
- As a result material becomes magnetized. The magnetization varies linearly with the strength of magnetic field. The induced dipole and magnetization vanish as soon as the applied field is switched off.

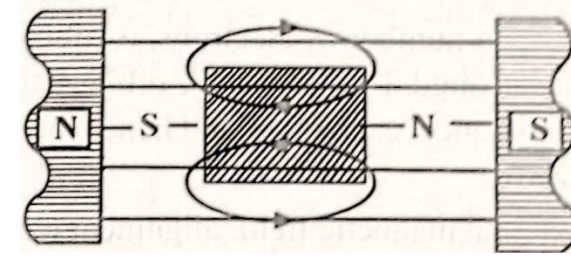


Characteristics of diamagnetic materials:

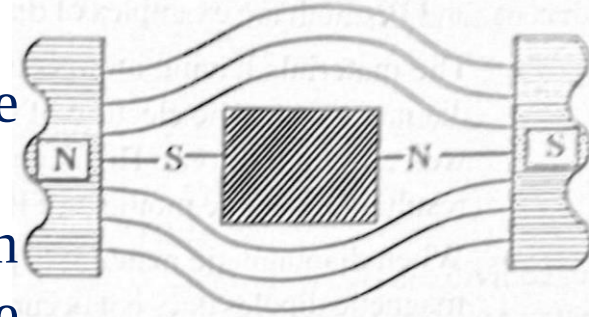
1. For diamagnetic materials, the magnetization M is directed opposite to the direction of applied magnetic field H . Hence the susceptibility is negative. The value of susceptibility is very small of the order of 10^{-6} .
2. Magnetic susceptibility is negative. hence the relative permeability μ_r is less than one ($\mu_r < 1$).
3. If a diamagnetic material is placed in an inhomogeneous magnetic field, it tends to be pushed into the region of weaker field.

Freely suspended bar of diamagnetic material will turn its direction perpendicular to the field lines. The field lines are pushed away from the material as shown in fig.

4. Magnetic susceptibility is independent of temperature. This was discovered by Pierre Currie in 1895.
5. The magnetization, M is a linear function of magnetic field H .
6. Solids, liquids and gaseous materials show diamagnetic behaviour.



a) magnetic field induced in a direction opposite to external field

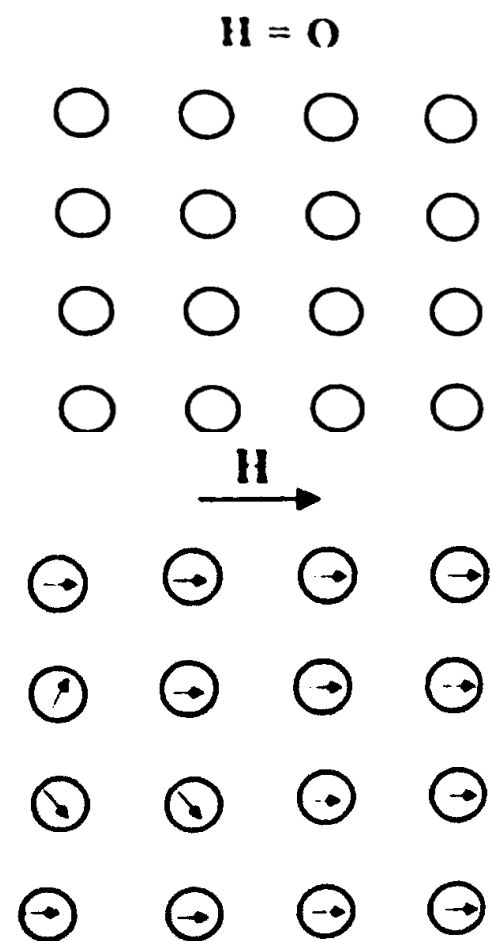


b) Magnetic field lines are pushed away from the diamagnetic material

Paramagnetic materials:

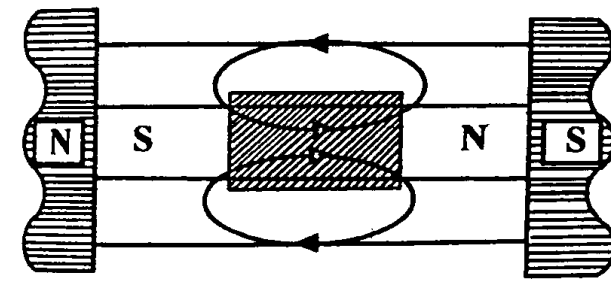
- Paramagnetic materials are the materials which when placed in a magnetic field acquire feeble magnetism in the direction of the magnetic field. e.g.: Oxygen. Copper chloride, Chromium. Platinum and solutions of Iron salts.
- Paramagnetic substance are freely attracted by magnets when placed in a magnetic field. They move from weaker to stronger parts of the field.
- Generally atoms and molecules of substances having odd number of electrons shows para-magnetism. They do not have total zero spin. Each atom or molecule possesses a permanent dipole moment by virtue of electron or orbital magnetic moments. In absence of an external magnetic field magnetic moments have random direction, hence no net macroscopic magnetic moments exist.

But when the magnetic field is applied, the dipole align themselves in the direction of applied field. The alignment of individual dipoles gives rise to net magnetization in the material as show in fig. The magnetic susceptibility of paramagnetic materials is 10^{-6} .

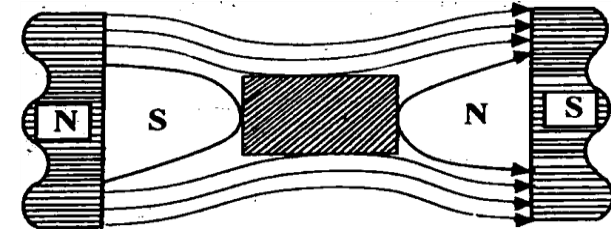


Characteristics of paramagnetic materials:

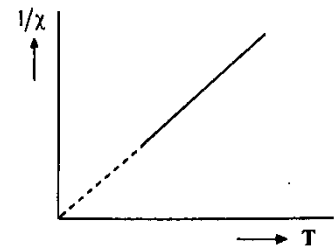
1. Paramagnetic materials exhibit positive magnetic susceptibility as the magnetization coincides the direction of magnetic field H .
2. The relative permeability is slightly more than one ($\mu_r > 1$). Magnetic field lines are pulled towards the material and also permeate through it when it is placed in magnetic field as shown in fig.
3. The magnetization (M) is linearly dependent on magnetic field (H).
4. Freely suspended rod of paramagnetic material aligns itself along the lines of induction. Paramagnetic substances are attracted towards stronger region of magnetic field. If a liquid is taken into a U shaped glass tube, placed in magnetic field then one leg of paramagnetic liquid rises in level as compared to other leg which is not under the action of magnetic field.
5. Paramagnetic susceptibility depends on temperature. Pierre Curie is discovered that the susceptibility of such material varies inversely with temperature as shown in fig.



Paramagnetic material gets magnetized in the direction of field.



Magnetic field lines are pulled towards the paramagnetic materials.



$$\chi = \frac{C}{T} \text{ ----- Curie law}$$

where C is called as curie constant. The above relation is called as Curie's law.

6. Solid, liquid and gaseous materials exhibit paramagnetic behavior.

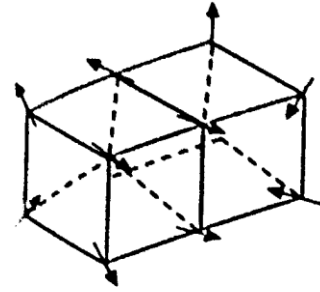
Ferromagnetic materials:

Materials when placed in magnetic field get strongly magnetized in the direction of the magnetic field are called as ferromagnetic materials.

e.g.: Iron, Nickel and some steels.

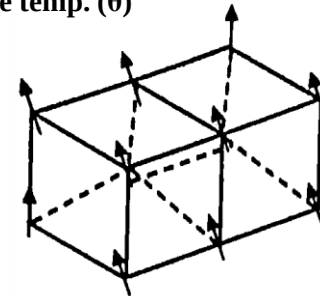
The experiments done by Einstein and de Hass show that the spin magnetic movement of electrons are responsible for ferromagnetism. Ferromagnetic materials exhibit spontaneous magnetization and a spontaneous magnetic movement even in the absence of applied magnetic field. Internal magnetic field induction is hundreds and thousands of times greater than induction of external magnetic field producing magnetization.

Fig. shows the orientation of spin movements above and below Curie temperature in absence of external magnetic field.



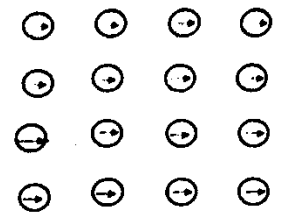
(a) $T > \theta$

Random orientation of spin movements in a ferromagnetic crystal above curie temp. (θ)



(b) $T < \theta$

Parallel alignment of spins at below curie temp. (θ)
 $H \propto I$

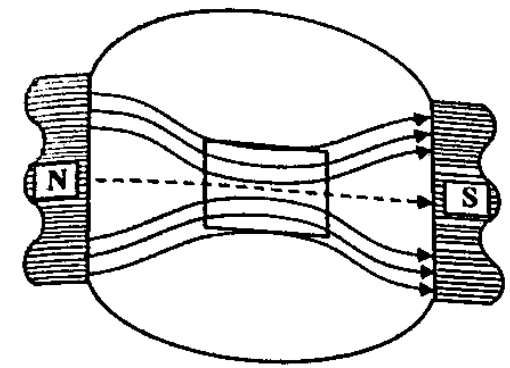


(b) $T < \theta$

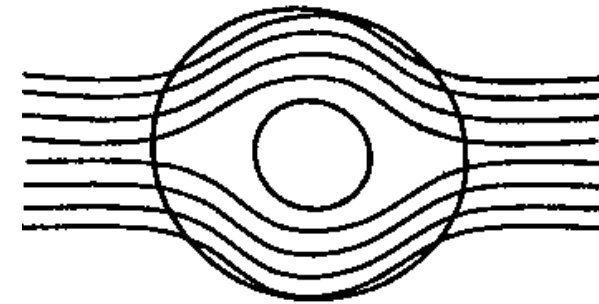
Alignment of spins in ferromagnetic material in the absence of external magnetic field below Curie temp. (θ)

Characteristics of ferromagnetic materials:

1. Ferromagnetic materials possess very high values of magnetic susceptibility of the order of 10^6 . Relative permeability is of the order of few thousand ($\mu_r \gg 1$).
2. If a ferromagnetic material is kept in a magnetic field, the field lines crowded into the material as shown in fig (a). If the material is taken in shell form, magnetic lines gets concentrated in the shell without penetrating into the cavity as shown in fig.(b). The material acts as a magnetic shield which does not allow the magnetic field to penetrate the space enclosed by the shell. Ferromagnetic materials conduct magnetic flux in a same way as metals conduct electric current.
3. Magnetization and magnetic field does not vary linearly in case of ferromagnetic material. It shows a complex nature. Fig 1 (a) shows the variation of H and B. As the magnetic field increases, initially magnetic induction (B) increases very fast. After certain value of H, magnetic saturation is reached and magnetization becomes independent of magnetic field. Curve OA shows the magnetization. This curve is known as initial magnetizing curve.



(a) Magnetic field lines are strongly pulled towards a ferromagnetic material.



(b) Magnetic shielding affected by a ferromagnetic shell.

Nonlinear relationship between H gives rise different values of permeability. Hence initial permeability μ and maximum permeability $\mu_{\max}$ calculated from OA curve is plotted as a function of H in fig. 1(b). Initially μ is increases and attains maximum value and then decreases rapidly as the value of H increases.

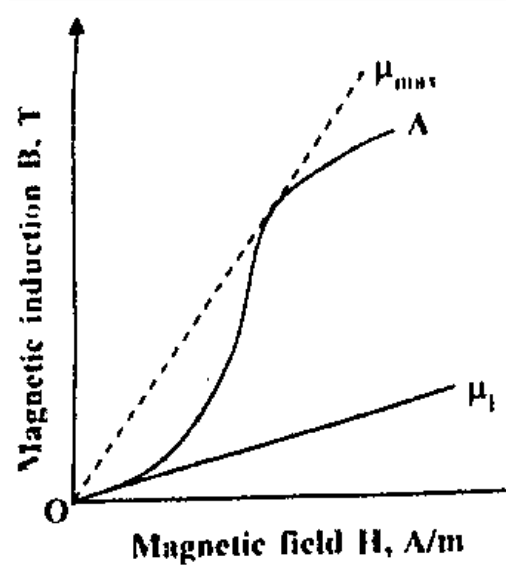


Fig. 1(a): Initial magnetization curve for ferromagnetic material slopes are drawn for $\mu_{\max}$ and μ .

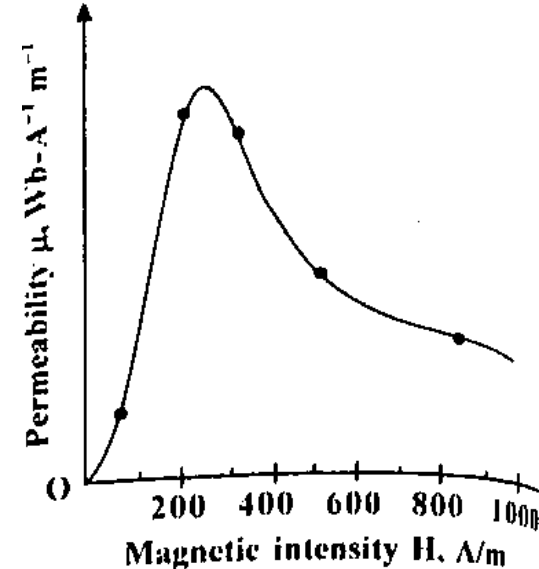
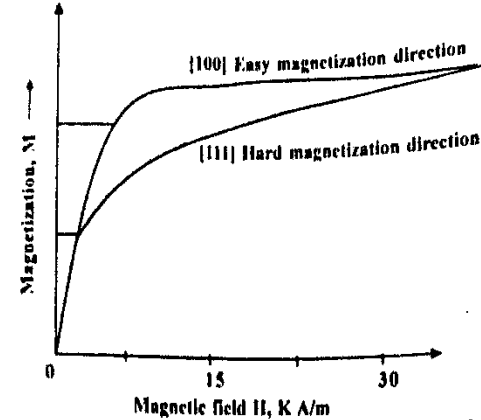


Fig. 1(b): The permeability μ of ferromagnetic material as a function of magnetic field H .

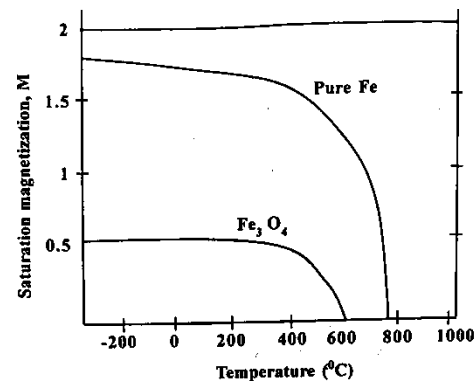
4. Specific property of ferromagnetic material shown only crystalline state temperature below Curie temperature. Diamagnetism and paramagnetism properties are displayed in any state of aggregation. Ferromagnetic material liquid and gaseous state behaves like paramagnetic substances.
5. Ferromagnetic properties of crystal are found be dependent the direction of magnetization.

In case of ferromagnetic materials, magnetization is obtained along crystallographic axes. The direction in which the magnetization is strong is called the direction of **easy magnetization**. The direction in which the magnetization is lowest it is called as **hard magnetization**. Iron has BCC structure. If the magnetic field is applied along [111] and [110] & [100], then the crystal easily magnetizes along [110] but it is not easy to magnetize along [111] body diagonal. Much higher field is required to magnetize it along this direction as shown in fig.



Relation of magnetic field H and magnetic induction in iron crystal along [111] & [100].

6. The ferromagnetic behavior of ferromagnetic materials disappears above curie temperature (T_c). The magnetization is maximum at 0°K and attains saturation. It decreases gradually with increasing temperature and drops to zero at curie temperature and vanishes beyond T_c and material transforms into a paramagnetic materials.



Plot of saturation magnetization versus temperature

Fig. shows the relation between temperature and saturation magnetization. The curie temperature T and the magnetic susceptibility of the paramagnetic state obey's Curie-Weiss Law.

$$\chi = \frac{C}{T - \theta} \quad T > T_c \quad \text{Where } C \text{ is curie constant and } \frac{1}{\chi}$$

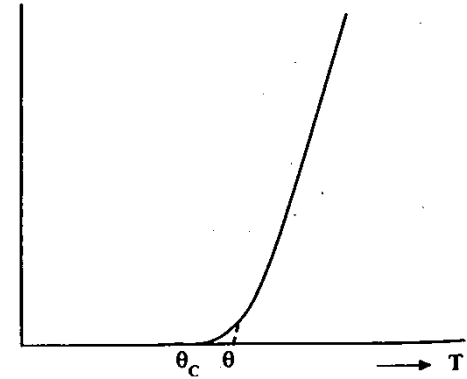
magnetization tends to infinity at $T = \theta$.

The variation of $1/\chi$ and temp. is shown in fig. The ferromagnetic behaviour was first noticed in iron.

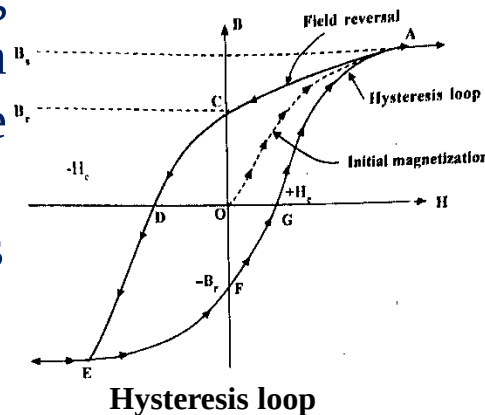
7. A typical property of ferromagnetic materials is hysteresis. Hysteresis is defined as the lag in changes of magnetization behind variation of the magnetic field. If, initially unmagnetized ferromagnetic material is subjected to increasing or decreasing magnetic fields, the magnetic field induction B varies as a function of H along a closed loop, called the hysteresis loop. The curve is shown in the fig.

The curve begins at 0. As the magnetic field H increases, B begins to increase slowly along the path OA . It attains a saturation value B_s and corresponding saturation magnetization M .

If H is decreased B also decreases but do not follow the same path. It follows path AC instead of AO . Thus B lags behind H . When H becomes zero, B does not become zero but has a value OC (B_r). The magnetic flux density remaining in the material is called the residual magnetism. This shows that the material remains magnetized even in the absence of applied field H . The power of retaining the magnetism is called the retentivity of the material.



A plot of reciprocal of susceptibility versus temp. for ferromagnetic material



Retentivity is a measure of the magnetic flux density remaining in the material when the magnetizing field is removed.

If the magnetic field is increased in the reverse direction the value of B decrease along the path CD and becomes zero. H attains value OD . Thus to reduce magnetic Induction a magnetic field of H must be applied in opposite direction to that of original magnetizing field. H is called as coercivity or the coercive force. It is a measure of the magnetic field strength required to destroy the residual magnetism in the material.

Further if H is increased in negative direction, saturation is attained in reverse direction denoted by point E in the above fig. A curve $FFGA$ is traced similar to $ACDE$ on reversing the field H Point C and F are such that the material is magnetized in the absence of external magnetic field. It is said to become permanent magnet. Closed curve $ACDEFGA$ is a cycle of magnetization and it is called as hysteresis loop.

Difference between paramagnetic, diamagnetic and ferromagnetic materials.

Sr. No.	Properties	Ferromagnetic material	Paramagnetic material	Diamagnetic material
1.	State	They are solids.	They can be solid, liquid or gas.	They can be solid, liquid or gas.
2.	Effect of magnet	Strongly attracted by a magnet	Weakly attracted by a magnet	Weakly repelled by a magnet.
3.	Behaviour in non-uniform magnetic field	Tends to move from low to high field region.	Tends to move from low to high field region.	Tends to move from high to low region.
4.	Behaviour under external field.	They preserve the magnetic properties after the external field is removed	They do not preserve the magnetic properties once the external field is removed.	They do not preserve the magnetic properties once the external field is removed.
5.	Susceptibility	Susceptibility χ is large and positive.	Susceptibility χ is less than one but positive.	Susceptibility χ is small but negative.
6.	Permeability	Very high.	Little greater than unity.	Little less than unity.
7.	Hysteresis	Exhibit phenomenon of hysteresis.	Does not exhibit hysteresis.	Does not exhibit hysteresis.
8.	Curie point	Have definite curie point above which they become paramagnetic.	No Curie point.	No Curie point.
9.	Retentivity	Possess retentivity.	No retentivity.	No retentivity.
10.	Examples	Iron, Steel, Cobalt, Nickel.	Platinum, chromium, Aluminium, salts of Fe, Ni, oxygen.	Bismuth, mercury, Silver, Copper, Water, Air.

Applications of Magnetic Materials:

1. Low Carbon Steel: Iron in pure state has high permeability and high electrical conductivity. This causes more eddy current losses. Low carbon steel (Fe 0.05% C) has small permeability and a higher resistivity. It is not expensive and hence used where low cost is important than other factors.

2. Iron-silicon Alloy's: They are prepared by adding 3-4% silicon to iron. Electrical resistivity of low carbon steel increases due to the addition of silicon. Thus eddy current losses can be reduced. It increases the magnetic permeability and lowers hysteresis losses. It also decreases magnetostriction and therefore reduces transformers noise, that is hum. Due to the addition of silicon, ductility tends to decrease and it makes the material brittle.

Grain orientation : Grain orientation plays important role in increasing permeability and decreasing hysteresis loss. In case of iron crystal [100] direction is easy direction and spin moments are aligned along this direction. When sheets of iron or iron alloys are manufactured by rolling and annealing, the direction [100] is parallel to rolling direction. Cold rolled grain orientation (CRGO) steels possess better magnetic properties in a direction parallel to the direction of rolling. Hence Cores require less material.

Many times aluminum and manganese of less than 1% are added in iron-silicon alloys to improve grain orientation and reduce hysteresis loss.

3. Nickel-iron Alloys:

For communication applications high permeability at low field is required. Nickel-iron alloys are used for these applications. These alloys are divided on the basis of nickel content in the alloy. They are 36% nickel, 50% nickel & 77% nickel.

- 36% nickel alloys have high resistivity and low permeabilities. They are used for high frequency devices such as speed relays, wideband transformers and inductors. They are cheap.
- 50% nickel alloys have moderate permeability ($\mu_{\max} = 25000$) and high saturation induction ($B_{\text{sat}} = 1.6 \text{ T}$). They are used where low loss and small size are required, such as relays, small motors and synchros.
- 77% nickel alloy has high permeability ($\mu_{\max} = 10^6$) but low saturation induction ($B_{\text{sat}} = 0.8 \text{ T}$). They are used in recording heads, pulse transformers, sensitive relays.

4. Alnico Alloys:

Permanent magnets are made using Alnico alloys. They contain aluminum, nickel, cobalt and iron with minor constituents such as copper & titanium. They have high energy product of range 40 to 70 KJ m³ and high remanent induction from 0.7 T to 13.5 T and moderate coercivity of 40 to 160 K A /m. Alnico alloys are hard and brittle. They are temperature resistant and shock proof.

5. Other Alloys:

- Rare earth alloys are superior than alnico alloys.
- Sm-Co alloys have energy products up to 240 KJ/m' and coercivities of about 3200 K A/m. They are used in medical devices such as thin motors and implantable pumps and valves.
- Alloys of magnetite Fe - Cr-Co are used in making permanent magnet for modern telephone receivers. They are developed in 1971.
- Hard Nd - Fe- B alloys have high energy products of the order of 300 KJ/m³. They are used in light weight and compact electric motors. They were discovered in 1984.

6. Magnetic storage:

- The semiconductor memories are volatile whereas the magnetic memories are permanent and non-volatile. The most common form of storage technology is magnetic storage.
- The data and programs required for computer operations are stored on magnetic bulk storage devices like tapes and disks. The first working magnetic recorder was invented by Valdemar Poulsen in 1898 and in 1928. Fritz Pfleumer developed the first magnetic tape recorders.

- The term magnetic storage is preferred in the field of audio and video production. The storage media is called a disk or cartridge. The process of storing data in a memory unit is called writing and process of retrieving data from memory is called reading. The medium used in magnetic storage devices is coated with iron-oxide. The oxide is ferromagnetic material.

In addition to the above applications magnetic materials have variety of applications in various fields such as

- Diamagnetic superconductors are used in magnetic levitation and making frictionless bearings.
- Paramagnetic iron oxide nanoparticles are used as contrast agents in magnetic resonance imaging.
- Ferromagnetic materials are used in magnetic recording devices and magnetic stripe on the back of credit card.
- Ferrimagnetic materials are used in high frequency applications.
- Ferrimagnetic materials are ceramic materials and are good electrical insulators.
- Magnetic nanoparticles are widely used in biomedical applications.
- Soft magnetic materials are used as sensors.
- Ceramic magnets have many engineering applications like Jet Milling, Vacuum Sintering, pressing, properties testing, vacuum melting etc.
- Permanent magnets are used as hard disk, floppy disk, CD driver head and in computer drivers.

List of Important Formulae:

$$1. \quad M = \chi H$$

$$3. \quad B = \mu_0 (H + M)$$

$$5. \quad \mu = \frac{B}{H} \quad , \quad \mu_0 = \frac{B_0}{H}$$

$$7. \quad \mu_r = 1 + \chi$$

$$9. \quad \chi_{\text{ferro}} = \frac{C}{T - \theta}$$

$$2. \quad B = \frac{\Phi}{A}$$

$$4. \quad B = \mu_0 (1 + \chi) H \quad , \quad M = H(\mu_r - 1)$$

$$6. \quad \mu_r = \frac{\mu}{\mu_0}$$

$$8. \quad \chi_{\text{para}} = \frac{C}{T}$$

Ex. 1 : A magnetic material has a magnetization of 2300 A/m and produces a flux density of 0.00314 Wb/m^2 . Calculate magnetizing force and relative permeability of the material.

Given :- $B = 0.00314 \text{ Wb/m}^2$, $M = 2300 \text{ A/m}$

$$\mu_0 = 4\pi \times 10^{-7} \text{ H/m} = 12.57 \times 10^{-7} \text{ H/m}$$

To find :- 1) Magnetizing force = H

2) Relative permeability = μ_r

Solution :

$$1) \quad H = \frac{B}{\mu_0} - M = \frac{0.00314 \text{ Wb/m}^2}{12.57 \times 10^{-7} \text{ H/m}} - 2300 \text{ A/m}$$

$$H = 198 \text{ A/m}$$

$$2) \quad \mu_r = \frac{B}{\mu_0 H} = \frac{0.00314 \text{ Wb/m}^2}{(12.57 \times 10^{-7} \text{ H/m}) \times (198 \text{ A/m})}$$

$$\mu_r = 12.61$$

Ex. 2: Diamagnetic Al_2O_3 is subjected to an external magnetic field of 10^5 A/m . Evaluate magnetization and magnetic flux density in Al_2O_3 (Susceptibility of $\text{Al}_2\text{O}_3 = -5 \times 10^{-5}$)

Given - $\chi = -5 \times 10^{-5}$, $H = 10^5 \text{ A/m}$

to find - 1) Magnetic flux density - B
2) Magnetization - M .

Solution :

$$1) M = \chi H$$

$$= (-5 \times 10^{-5}) \times (10^5 \text{ A/m})$$

$$M = -5 \text{ A/m}$$

$$2) B = \mu_0 (H + M)$$

$$= (12.57 \times 10^{-7} \text{ H/m}) (10^5 \text{ A/m} - 5 \text{ A/m})$$

$$B = 0.126 \text{ Wb/m}^2$$

Ex. 3: The susceptibility of paramagnetic FeCl_3 is 3.7×10^{-3} at 27°C . What will be the value of its susceptibility at 200°K and 500°K ?

Given :- $T = 200\text{ K}$ and $T = 500\text{ K}$

To find :- $\chi_{200\text{ K}}$ and $\chi_{500\text{ K}}$ that is susceptibility at 200 K and 500 K .

Solution :

$$1) \chi = \frac{C}{T}$$

$$C = \chi T = (3.7 \times 10^{-3}) \times 300\text{ K} \\ = 1.11\text{ K}$$

$$\chi_{200\text{ K}} = \frac{C}{T} = \frac{1.11\text{ K}}{200\text{ K}}$$

$$\boxed{\chi_{200\text{ K}} = 5.55 \times 10^{-3}}$$

$$2) \chi_{500\text{ K}} = \frac{C}{T} = \frac{1.11\text{ K}}{500\text{ K}}$$

$$\boxed{\chi_{500\text{ K}} = 2.22 \times 10^{-3}}$$

Ex. 4 : Find the relative permeability of the ferromagnetic material if a magnetic field of strength 220 A/m produces magnetization of 3300 A/m in it.

Given :- $M = 3300 \text{ A/m}$, $H = 220 \text{ A/m}$

To find :- Relative permeability = μ_r

Solution :

$$\mu_r = 1 + \chi$$

$$= 1 + \frac{M}{H}$$

$$= 1 + \frac{3300 \text{ A / m}}{220 \text{ A / m}}$$

$$\mu_r = 16$$

Ex. 5 : The magnetic susceptibility of silicon is -0.4×10^{-5} . Calculate the flux density and magnetic moment per unit volume when magnetic field of intensity $5 \times 10^5 \text{ A / m}$ is applied.

Given :- $\chi = -0.4 \times 10^{-5}$, $H = 5 \times 10^5 \text{ A / m}$

To find :- (1) B = magnetic flux density

(2) magnetic moment per unit volume = M

Solution :-

$$1) \quad B = \mu_0 (H + M)$$

$$= \mu_0 H(1 + \chi)$$

$$= 4\pi \times 10^{-7} \times 5 \times 10^5 (1 + (-0.4) \times 10^{-5})$$

$$\boxed{B = 0.62 \text{ Wb / m}^2}$$

$$2) \quad M = \chi H$$

$$= -0.4 \times 10^{-5} \times 5 \times 10^5$$

$$\boxed{M = -2.0 \text{ A / m}}$$

Ex. 6 : The magnetic field strength in silicon is 1000 A/m . If the magnetic susceptibility is -0.25×10^{-5} , Calculate the magnetization and flux density in silicon.

Given :- $H = 1000 \text{ A/m}$, $\chi = -0.25 \times 10^{-5}$

To find :- Magnetization = M and B = Magnetic flux density

Solution :

$$1) \quad M = \chi H$$

$$= -0.25 \times 10^{-5} \times 1000 \text{ A / m}$$

$$\boxed{M = -0.25 \times 10^{-2} \text{ A / m}}$$

$$2) \quad B = \mu_0 (H + M)$$

$$= 4\pi \times 10^{-7} (1000 - 0.25 \times 10^{-2})$$

$$\boxed{B = 1.25 \times 10^{-3} \text{ Wb / m}^2}$$

Ex. 7: When a magnetic material is subjected to a magnetic field of intensity 250 A/m. Its relative permeability is 15. Calculate its magnetization and magnetic flux density.

Given :- $H = 250 \text{ A/m}$, $\mu_r = 15$

To find :- Magnetization = M and Magnetic flux density = B

Solution :

$$\begin{aligned} 1) \quad M &= H(\mu_r - 1) \\ &= 250 \text{ A / m } (15 - 1) \\ &= 250 \times 14 \end{aligned}$$

$$\boxed{M = 3500 \text{ A / m}}$$

$$\begin{aligned} 2) \quad B &= \mu_0 (H + M) \\ &= 4\pi \times 10^{-7} (250 \text{ A / m}) + 3500 \text{ A / m} \end{aligned}$$

$$\boxed{B = 4.71 \times 10^{-3} \text{ Wb / m}^2}$$

Ex. 8 : A magnetic material has a magnetization of 3300 A/m and flux density of 0.0044 Wb/m². Calculate the magnetizing force and the relative permeability of the material.

Given :- Magnetization, $M = 3300 \text{ A/m}$, $B = 0.0044 \text{ Wb/m}^2$

To find :- Magnetizing force = H and relative permeability = μ_r

Solution :

$$1) \quad B = \mu_0 (H + M)$$

$$\therefore H = \frac{B}{\mu_0} - M$$

$$= \frac{0.0044 \text{ Wb/m}^2}{4 \times \pi \times 10^{-7}} - 3300 \text{ A/m}$$

$$H = 203 \text{ A/m}$$

$$2) \quad M = H(\mu_r - 1)$$

$$\mu_r = 1 + \frac{M}{H}$$

$$= 1 + 165$$

$$\boxed{\mu_r = 17256}$$

Ex. 9 : A paramagnetic material has 10^{28} atom per m^3 . Its susceptibility at 350 K is 2.8×10^{-4} . Calculate the susceptibility at 300 K.

Given :- $N = 10^{28}$ atoms/ m^3 , Susceptibility at T_1 , $\chi_1 = 2.8 \times 10^{-4}$ where $T_1 = 350$ K

To find :- Susceptibility at T_2 , χ_2 where $T_2 = 300$ K

Solution : For paramagnetic material

$$\chi = \frac{C}{T} \quad \text{or} \quad C = \chi T$$

$$\therefore \chi_1 T_1 = \chi_2 T_2$$

$$\chi_2 = \frac{\chi_1 T_1}{T_2}$$

$$= \frac{2.8 \times 10^{-4} \times 350}{300}$$

$$\boxed{\chi_2 = 3.267 \times 10^{-4}}$$

Ex. 10 : The magnetic field in the interior of a certain solenoid has the value of $6.5 \times 10^{-4} \text{ T}$, when the solenoid is empty. When it is filled with iron, the field becomes 1.4 T. Find the

relative permeability of iron.

Given :- $B_0 = 6.5 \times 10^{-4} \text{ T}$

$$B = 1.4 \text{ T}$$

To find :- Relative permeability of iron = μ_r

Solution :

$$\begin{aligned}\mu_r &= \frac{B}{B_0} \\ &= \frac{1.4 \text{ T}}{6.5 \times 10^{-4} \text{ T}}\end{aligned}$$

$$\boxed{\mu_r = 2153.85}$$